

# Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: AFRIDOC\_MT | Context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

AFRIDOC-MT is a ground-up document-level MT benchmark designed for African languages including Amharic, with a health domain category that aligns with the deployment's prose-narrative scope [Q1, Q2, Q15]. The user has explicitly scoped structured document types (tables, timetables, forms) as out-of-scope, which mitigates the most obvious category gap. However, the benchmark mixes a tech-news category (Techpoint Africa) [Q18, Q22] that introduces construct-irrelevant variance for clinical deployments, and within the health category a substantial portion of sampled documents covers topics geographically or substantively distant from Ethiopian MOH clinical priorities (Afghanistan polio, CCHF, volcanoes, Zika, TRIPS policy) [DATASET-D7, D32, D33, D34, D26, D46]. Multi-way translation across 30 directions is supported but the deployment uses only en→am [Q7, Q94, Q101]. Net: the taxonomy partially aligns — document-level MT for Amharic prose health is covered, but the benchmark's category structure does not isolate clinical-guideline / maternal-health / vaccination subgenres that the deployment actually translates.

| ID   | Evidence                                                                                                                                                                                                                                                                                          |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1   | "This paper introduces AFRIDOC-MT, a document-level multi-parallel translation dataset covering English and five African languages: Amharic, Hausa, Swahili, Yorùbá, and Zulu." (p.1)                                                                                                             |
| Q2   | "The dataset comprises 334 health and 271 information technology news documents, all human-translated from English to these languages." (p.1)                                                                                                                                                     |
| Q7   | "In addition, AFRIDOC-MT supports multi-way translation, allowing translations not only between English and the African languages but also between any two of the languages covered." (p.1)                                                                                                       |
| Q15  | "Document-level NMT aims to overcome the limitations of sentence-level systems by translating an entire document as a whole." (p.2)                                                                                                                                                               |
| Q18  | "We scraped English articles from the websites of Techpoint Africa and the World Health Organization (WHO). The articles cover different topics of different lengths with an average length of 30 and 37 sentences for health and tech respectively." (p.3)                                       |
| Q22  | "Finally, we selected 334 and 271 English articles/documents from the health and tech domains respectively, which represents 10k sentences each per domain." (p.3)                                                                                                                                |
| Q94  | "AFRIDOC-MT is a multi-way parallel dataset. However, due to the cost of running inference over three prompts and across all 30 translation directions for all the models evaluated, most of our analysis is limited to translation tasks between English and the five African languages." (p.10) |
| Q101 | "While we fine-tuned both NLLB-1.3B and LLaMAX3 models across all 30 language directions, due to computational constraints and the high cost of qualitative evaluation, our detailed analysis focuses only on translation between English and the 5 African languages." (p.10)                    |
| D7   | ነገር ግን፣ በ2023 እስከ ዛሬ 46 ደብሊዊፒቪ1 አዎንታዊ የአካባቢ ናሙናዎች ... ናንጋርሃር እና ኩናር አውራጃዎች ... ሚሊዮን ተመላሾች — Amharic WHO article on Afghanistan polio — geographic specificity to Afghanistan/Pakistan, not Ethiopia; pseudo-document chunk (10 sentences)                                                         |
| D26  | ዚካ ቫይረስ ... ፀረ-ተባይ ... Vector control operations framework for Zika virus — Short Zika virus sentence fragment — topic not endemic to Ethiopia                                                                                                                                                    |
| D32  | Crimean-Congo haemorrhagic fever (CCHF) is a viral haemorrhagic fever usually transmitted by ticks ... CCHF is endemic in all of Africa, the Balkans, the Middle East and in Asia. — CCHF article — hemorrhagic fever not a primary health priority in Ethiopia; tick-borne virus                 |
| D33  | A volcano is a vent in the Earth's crust from which eruptions occur... There are about 1500 potentially active volcanoes worldwide. — Volcanic eruption WHO article — not relevant to Ethiopian clinical translation deployment                                                                   |

|            |                                                                                                                                                                                                                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D34</b> | Afghanistan continues to implement an intense campaign schedule, focusing on improving quality in the endemic zone... massive population movement significantly increases the risk of cross-border poliovirus spread — WHO Afghanistan polio article — geographically specific to Afghanistan, not Ethiopia |
| <b>D46</b> | TRIPS Agreement, many resolutions of the World Health Assemblies have requested WHO to address the impact of trade agreements and intellectual property protection on public health and access to medicines — TRIPS/IP health policy — highly technical policy content                                      |

## Strengths

- Ground-up design for African languages including Amharic with document-level MT as a distinct construct from sentence-level MT [Q1, Q15, Q16]
- Health domain category is present and operationalized at document level — the closest existing category match for the deployment [Q2, Q14, DATASET-D4, D15, D42]
- Deployment scope (prose health) aligns with the benchmark's narrative-prose-only corpus structure [Q18, DATASET-D1, D3]

| ID         | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1</b>  | "This paper introduces AFRIDOC-MT, a document-level multi-parallel translation dataset covering English and five African languages: Amharic, Hausa, Swahili, Yorùbá, and Zulu." (p.1)                                                                                                                                                                                                                                                                              |
| <b>Q2</b>  | "The dataset comprises 334 health and 271 information technology news documents, all human-translated from English to these languages." (p.1)                                                                                                                                                                                                                                                                                                                      |
| <b>Q14</b> | "To the best of our knowledge, only TICO-19, a health domain translation benchmark, has the potential to be used for document-level MT, while it is restricted to topics related to COVID-19." (p.2)                                                                                                                                                                                                                                                               |
| <b>Q15</b> | "Document-level NMT aims to overcome the limitations of sentence-level systems by translating an entire document as a whole." (p.2)                                                                                                                                                                                                                                                                                                                                |
| <b>Q16</b> | "Both document-level and context-aware MT allow for the possibility of improving translation quality for context-dependent phenomena such as coreference resolution (Müller et al., 2018; Bawden et al., 2018; Voita et al., 2018; Herold and Ney, 2023), lexical disambiguation (Rios Gonzales et al., 2017; Martínez Garcia et al., 2019), and lexical cohesion (Wong and Kit, 2012; Garcia et al., 2014, 2017; Bawden et al., 2018; Voita et al., 2019)." (p.2) |
| <b>Q18</b> | "We scraped English articles from the websites of Techpoint Africa and the World Health Organization (WHO). The articles cover different topics of different lengths with an average length of 30 and 37 sentences for health and tech respectively." (p.3)                                                                                                                                                                                                        |
| <b>D1</b>  | የሞቃታማ ዐውሎ ነፋሶች ... ለሞቃታማ አውሎ ነፋሶች፣ እንዲሁም ሀይለኛ አውሎ ነፋሶች ወይም አውሎ ነፋሶች ... ኃይለኛ ክብ አውሎ ነፋሶች ... 233 000 ሰዎችን ገድለዋል — Full Amharic document on tropical cyclones (WHO article), multi-sentence, Ethiopic script, with clinical/epidemiological content and specific numbers                                                                                                                                                                                            |
| <b>D3</b>  | አካል ጉዳተኝነት ሰው የመሆን አካል ነው። ሁሉም ሰው ማለት ይቻላል በሕይወታቸው ላይ በሆነ ወቅት ጊዜያዊ ወይም ዘላቂ የአካል ጉዳት ያጋጥማቸዋል — Amharic WHO article on disability — prose narrative, full document preserved                                                                                                                                                                                                                                                                                         |
| <b>D4</b>  | ተጨማሪ ምግብ ማጠቃለያ፡- በ 6 ወር አካባቢ የሕፃን የኃይል ፍላጎት ... ጨቅላ ህፃናት ... ጡት ማጥባት — Amharic complementary feeding article — maternal-child health topic directly relevant to deployment (maternal-health booklets)                                                                                                                                                                                                                                                              |
| <b>D15</b> | ደህንነቱ የተጠበቀ፡- በንጽህና ተከማችተው እና ተዘጋጅተዋል እንዲሁም ጠርመሶች እና ፕላስቲኮችን ሳይሆን በንጹህ እቃዎች ... ጡት ማጥባት — Complementary feeding clinical guidance in Amharic — high relevance (maternal health booklet content)                                                                                                                                                                                                                                                                    |
| <b>D42</b> | የዓለም ጤና ድርጅት ሕፃናትን ለ6 ወራት ብቻ እንዲያጠቡ ይመክራል ... ደህንነቱ የተጠበቀ እና ተጨማሪ ምግብ — Amharic WHO recommendation on exclusive breastfeeding — direct match for maternal-health booklet content                                                                                                                                                                                                                                                                                   |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                            |
|-------------|----------------------------------------------------------------|
| <b>IO-1</b> | Identify test case categories required for regional deployment |
| <b>IO-2</b> | Check if taxonomy omits regionally relevant categories         |
| <b>IO-3</b> | Check if taxonomy includes irrelevant categories               |
| <b>IO-4</b> | Document category gaps harming content validity                |

## Checklist responses:

**IO-1:** Required categories for this deployment are clinical practice guidelines (narrative), maternal-health booklets, and vaccination schedule narratives — all prose. Benchmark provides a 'health' category from WHO articles that partially covers these subgenres [Q2, Q18, DATASET-D4, D9, D42].

**IO-2:** The benchmark does not subdivide health into clinical-guideline / maternal-health / immunization-schedule / endemic-disease subcategories. Ethiopia-specific subgenres (e.g., MOH referral pathways, endemic-disease guidance) are absent — only TICO-19 is mentioned as a comparable health document-level resource and is COVID-restricted [Q14]. The deployment-relevant maternal-child / immunization subset is present but constitutes a minority of sampled health documents [DATASET-D4, D15, D42, D31, D44].

**IO-3:** The benchmark includes a tech-news category (Techpoint Africa, 271 documents) that is irrelevant to clinical translation deployment [Q2, Q18, DATASET-D13, D14, D22, D38, D39]. Within health, a substantial share of sampled WHO documents covers globally generic or non-Ethiopian-relevant topics (Afghanistan polio, CCHF, volcanoes, TRIPS, IHR, Zika) [DATASET-D7, D32, D33, D34, D46, D26].

**IO-4:** Two category-level issues harm content validity: (a) inclusion of irrelevant tech-domain category that dilutes any aggregate score [DATASET-D14, D22], and (b) absence of fine-grained clinical-guideline / maternal-health / endemic-disease subcategorization within the health domain, meaning a system can score well on aggregate health while underperforming on the deployment-specific subgenres [DATASET-D7, D33].

| ID  | Evidence                                                                                                                                                                                                                                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q2  | "The dataset comprises 334 health and 271 information technology news documents, all human-translated from English to these languages." (p.1)                                                                                                                                                               |
| Q14 | "To the best of our knowledge, only TICO-19, a health domain translation benchmark, has the potential to be used for document-level MT, while it is restricted to topics related to COVID-19." (p.2)                                                                                                        |
| Q18 | "We scraped English articles from the websites of Techpoint Africa and the World Health Organization (WHO). The articles cover different topics of different lengths with an average length of 30 and 37 sentences for health and tech respectively." (p.3)                                                 |
| D4  | ተጨማሪ ምግብ ማጠቃለያ፡- በ 6 ወር አካባቢ የሕፃኑ የኃይል ፍላጎት ... ጨቅላ ህፃናት ... ጡት ማጥጣት — Amharic complementary feeding article — maternal-child health topic directly relevant to deployment (maternal-health booklets)                                                                                                       |
| D7  | ነገር ግን፡ በ2023 እስከ ዛሬ 46 ደብሊዊፒቪ1 አዎንታዊ የአካባቢ ናሙናዎች ... ናንጋርሃር እና ኩናር አውራጃዎች ... ሚሊዮን ተመላሾች — Amharic WHO article on Afghanistan polio — geographic specificity to Afghanistan/Pakistan, not Ethiopia; pseudo-document chunk (10 sentences)                                                                   |
| D9  | ዓለም አቀፍ ትብብር የጋራ ዓላማን ለማሳካት ... ዜጅ ዶዝ ... ህፃናት ፈልጎ ማግኘት ... ዶ/ር ጆርጅ ምዊንያ — Amharic WHO Assembly article about immunization — broad vaccination language relevant to vaccination schedule booklets                                                                                                           |
| D13 | ፔይ ዲይ ለሽያጭ ? ዛሬ በ ቴክ ፖይንት አፍሪካ ... የየይጂሪያ ፕረዚዳንት ... ፔይ ዲይ ... ኦባሲ እነ-አቦንግ — Amharic Techpoint Africa tech-news article about Nigerian fintech/genomics — not relevant to Ethiopian MOH deployment                                                                                                          |
| D14 | ደቡብ አፍሪካ ጎጂ ይዘቶችን በመስመር ላይ ማቆም ትፈልጋለች ... YouTube ... NALA ... Cellulant ... ሴሉላንት 20% — Long Amharic tech article: South Africa content regulation, YouTube, NALA remittances, Cellulant layoffs — entirely irrelevant to Ethiopian clinical translation                                                   |
| D15 | ደህንነቱ የተጠበቀ፡- በንጽህና ተከማችተው እና ተዘጋጅተዋል እንዲሁም ጠርመሶች እና ፕላስቲኮችን ሳይሆን በንጹህ እቃዎች ... ጡት ማጥጣት — Complementary feeding clinical guidance in Amharic — high relevance (maternal health booklet content)                                                                                                             |
| D22 | Fatanmi identifies its effect on marketing a business ... Gen Zs ... Building in public... — Tech article about startup marketing strategy — entirely irrelevant to Ethiopian MOH clinical translation deployment                                                                                           |
| D26 | ዚካ ቫይረስ ... ፀረ-ተባይ ... Vector control operations framework for Zika virus — Short Zika virus sentence fragment — topic not endemic to Ethiopia                                                                                                                                                              |
| D31 | ጡት ማጥጣት ... ሞተዋል ... 820,000 ... 40% ... — Amharic infant nutrition WHO article — directly relevant to maternal-health booklet                                                                                                                                                                              |
| D32 | Crimean-Congo haemorrhagic fever (CCHF) is a viral haemorrhagic fever usually transmitted by ticks ... CCHF is endemic in all of Africa, the Balkans, the Middle East and in Asia. — CCHF article — hemorrhagic fever not a primary health priority in Ethiopia; tick-borne virus                           |
| D33 | A volcano is a vent in the Earth's crust from which eruptions occur... There are about 1500 potentially active volcanoes worldwide. — Volcanic eruption WHO article — not relevant to Ethiopian clinical translation deployment                                                                             |
| D34 | Afghanistan continues to implement an intense campaign schedule, focusing on improving quality in the endemic zone... massive population movement significantly increases the risk of cross-border poliovirus spread — WHO Afghanistan polio article — geographically specific to Afghanistan, not Ethiopia |
| D38 | የሕንድ ኢድቴክ ኢንዱስትሪ ... ሕንድ ከዩናይትድ ስቴትስ (ዩ ኤስ) ቀጥሎ ... edtech ... Central Square Foundation ... Nigerian Edtech ecosystem — Indian/Nigerian edtech article — entirely irrelevant to Ethiopian clinical translation                                                                                             |

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|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D39</b>    | Nigerian healthtech startup, Clafiya, raises \$610,000 in pre-seed funding — African healthtech/fintech startup news — entirely irrelevant to Ethiopian MOH clinical translation deployment                                                                            |
| <b>D42</b>    | የዓለም ጤና ድርጅት ሕፃናትን ለ6 ወራት ብቻ እንዲያጠቡ ይመክራል ... ደህንነቱ የተጠበቀ እና ተጨማሪ ምግብ — Amharic WHO recommendation on exclusive breastfeeding — direct match for maternal-health booklet content                                                                                       |
| <b>D44</b>    | ጨቅላ ህፃናት ከ6 ወር ጀምሮ የተጣራ፣ የተፈጨ እና ከፊል ጠጣር ምግቦችን መመገብ ይችላሉ — Amharic instruction for pureed/mashed foods from 6 months — clinically precise maternal-health instruction                                                                                                  |
| <b>D46</b>    | TRIPS Agreement, many resolutions of the World Health Assemblies have requested WHO to address the impact of trade agreements and intellectual property protection on public health and access to medicines — TRIPS/IP health policy — highly technical policy content |
| <b>WEB-10</b> | JMIR 2025 scoping review identified only 3 Amharic NLP-for-public-health studies; none evaluated Ethiopia-endemic disease terminology ( <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC11923465/">https://pmc.ncbi.nlm.nih.gov/articles/PMC11923465/</a> )          |

## Information Gaps

- Proportion of WHO health documents that are clinical-guideline / maternal-health / immunization vs. policy / financing / non-Ethiopian-endemic — not quantified beyond 51-document sample
- Whether vaccination schedule narrative (a deployment subgenre) is meaningfully present in the 334-document health corpus

## Requires Expert Verification

- Whether the AFRIDOC-MT health subgenre distribution adequately represents the document types Ethiopia MOH actually distributes (requires comparison with MOH document inventory)